Building and Environment 171 (2020) 106675

Contents lists available at ScienceDirect

Building and Environment

journal homepage: http://www.elsevier.com/locate/buildenv

Greening the classroom: Three field experiments on the effects of indoor nature on students' attention, well-being, and perceived environmental quality

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ARTICLE INFO

Keywords:
Restorative environments
Wellbeing
Potted plants
Green space
Classroom

ABSTRACT

Prior work has shown beneficial effects of indoor nature (e.g. potted plants, green walls) on attention, health, and well-being for various groups, including students. The aim of this paper was to examine whether these effects also emerge when students attend a single lecture in a classroom with indoor nature. Three longitudinal field experiments, conducted at a university (N = 70), secondary school (N = 213), and a secondary vocational school (N = 161), examined students' attention, well-being, health complaints, lecture evaluation, and perceived environmental quality with attentional tests and questionnaires. Perceived environmental quality of classrooms with (rather than without) indoor nature was consistently rated more favourably. Secondary education students also reported greater attention, lecture evaluation, and teacher evaluation after one lecture in classrooms with indoor nature compared to the classroom without. There were no straightforward intervention effects on well-being and health complaints. Although this research provides some support for indoor nature in classrooms, attending only one lecture in a classroom with indoor nature does not seem to provide immediate effects on health and well-being.

1. Introduction

Students in high schools and universities often experience study-related stress due to high expectations from themselves and others, pressures from exams and classes, and lack of time, skills, money, sleep, or support [1]. These and related stressors can have a negative impact on students' health, well-being, and academic achievement [2–5]. During their time in classrooms, students are required to focus, absorb information, and actively reflect on this information. These tasks call upon students' directed attention resources, which are susceptible to fatigue. This mental fatigue can lead to a reduced ability to solve problems, difficulties with inhibition, an increase in distractions, errors, irritable feelings, and susceptibility to stress [6,7]. If students experience such negative feelings during their lectures, these feelings may persist during the day and may influence their overall well-being. To help students cope with demands of everyday life, attention is being paid to how the physical classroom environment can serve as a supportive environment that boosts and protects students' overall well-being and performance [8–10]. One promising intervention in this regard is the introduction of indoor nature (e.g. potted plants, green walls) in the classroom [11,12].

1.1. The case of indoor nature

Indoor nature can provide a range of positive physical and psychological health benefits [11,13]. These benefits may be explained by two complementary mechanisms: a psychological mechanism and a physiological mechanism. First, the psychological mechanism is based on an evolutionary perspective. The biophilia hypothesis states that all humans innately feel attracted, have affection for, and respond positively to natural elements and life forms [14]. The biophilia hypothesis is stated in general terms, which means that it applies to a variety of domains. As such, biophilia may also explain people's positive response to indoor nature. In line with this, people have been found to report improved evaluations of office settings or hospital rooms after plants

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https://doi.org/10.1016/j.buildenv.2020.106675
Received 11 June 2019; Received in revised form 10 January 2020; Accepted 11 January 2020
Available online 22 January 2020
0360-1323/

were placed [15–18]. Building from this evolutionary assumption, nature is also believed to provide a breather from feelings of stress and mental fatigue, or so called restorative experiences [19]. Work on psychological restoration has generally been guided by the attention restoration theory [6,20] and stress recovery theory [21,22]. Attention restoration theory suggests that nature may engage attention in an effortless manner, which allows the directed attention resources to rest and restore [6,20]. Stress recovery theory proposes that interacting with nature elicits a positive affect response that in turn effects physical and psychological functioning related to stress reduction and positively-toned emotions [21,22]. In line with these theories, previous studies in office settings have shown that indoor nature is associated with greater productivity [12,17,23], less stress [24], and more positive emotional well-being [25].

Second, the physiological mechanism lies in nature's ability to contribute to the indoor climate. In classrooms, large numbers of people in a confined and limited space in combination with poor ventilation, electronic equipment, and many other factors can result in low humidity and rising temperatures, carbon dioxide concentrations, level of volatile organic compounds, and other biological agents [26,27]. This is concerning because poor indoor climate may reduce feelings of comfort and has been associated with reduced attention, vigilance, study performance, and more health complaints such as fatigue, headaches, irritation of eyes, nose, and throat, and nausea in teachers and students [28–34]. Indoor nature, such as plants and green walls, can help increase the relative humidity, and reduce temperature, carbon dioxide concentration, and level of volatile organic compounds [35–39].

1.2. *Indoor nature in the classroom*

The theoretical framework of Abdelaal [40] proposes that including biophilic elements such as plants, natural landscapes, light, and water in learning spaces, such as classrooms, supports its learning function and can thereby generate a sustainable, inspiring, and innovating learning environment. Previous empirical work has already explored the effects of indoor nature in primary school classrooms. Those studies have shown beneficial effects of indoor nature on pupils attention [41], academic performances [42], rates of misbehaviours, hours of sick leave [43], and health complaints related to the indoor climate [44]. For example, van den Berg et al. [41] showed that pupils scored better on a selective-attention task and rated the attractiveness of the classroom higher in classrooms with green walls than those without. Given these encouraging results, it is important to test whether these effects hold in classrooms where students attend a single lecture instead of several hours per day (on most weekdays) such as in the case of primary schools. High school and university students (i.e., secondary and tertiary education students) generally follow several 1-h lectures in different classrooms during the day. Each individual lecture calls upon students' adaptive resources, and the physical classroom environment may play a role in this. In this present study, we examined if following one lecture in a classroom with indoor nature influences students' overall health and well-being.

To the best of our knowledge, three studies have examined the effects of indoor nature in classrooms were students attend lectures for only a short period. First, a post-intervention questionnaire study showed that Pakistani university students reported improved indoor climate and academic performance after plants were placed in their classrooms [45]. Second, university students in the United States who followed a psychology course in a classroom with plants evaluated the course and the professor at the end of the semester more positively than those who took the same course in a standard classroom [46]. Third, using a controlled experimental design, knowledge retention of undergraduate students in the United Kingdom was higher after a short lecture in a classroom with natural elements (window views, plants, nature scent) than in a lecture hall lacking these elements [47]. There is thus initial evidence that indoor nature in classrooms where students attend one lecture could contribute to students' perceptions and cognitive functioning. However, it remains unclear if this short exposure can also induce immediate beneficial effects on students' health and well-being.

1.3. *The present research and hypotheses*

The objective of the three field experiments described in this paper was to extend previous findings by investigating the effects of indoor nature in classrooms on students' attention, well-being, health complaints, teacher and lecture evaluation, and perceived environmental quality after one lecture. We hypothesised that indoor nature in classroom would improve students' attention, well-being, health, teacher and lecture evaluations, and perceived environmental quality. The studies were conducted in real classroom settings at three different education institutes in the Netherlands. Examining these different settings allowed us to explore if the results could be generalised across educational settings.

Our objective was to add-on new information with each study. In Study 1, we only examined the "short-term" effects of indoor nature (i.e., shortly after integration of indoor nature). In the Study 2 and 3, we used longitudinal designs to explore if the potential effects were a result of novelty or if they persisted over the longer term (i.e., few weeks after integration of indoor nature). Additionally, in Study 1 indoor nature was conceptualized using both plants and green walls. In Study 2 and 3, we included one sort of indoor nature per classroom trying to unravel differences in effects. In Study 2, a flower only condition was included because flowers are generally more colourful and have more scent than plants. We were, in an explorative manner, interested if these aspects would also contribute to students' classroom experiences. In Study 3, we included a green walls only condition because if they are proven effective, they may be a low maintenance solution to bring indoor nature to classrooms.

2. Study 1

2.1. *Method*

2.1.1. *Design and procedure*

The study included a classroom with the standard design (control) and two different classrooms with indoor nature that together represent the experimental condition. There were two measurements (T1 and T2), both on a Wednesday morning at 11:45 one week apart during two obligatory lectures on Research and Statistics in April 2016. The study thus had a two (classroom condition; between subject) by two (measurement; within subjects) design, and included elements of a crossover and a parallel design. Two groups followed a crossover design, with a different exposure (control or indoor nature) at T1 and T2. The third group was exposed to indoor nature at T1 and T2 (Table 1).

Table 1
Characteristics of students included in Study 1, Study 2, and Study 3.

	n	Female	Age
		% (n)	Mean (*SD*)
Study 1 – university	**70**	**88.6 (62)**	**19.5 (1.4)**
Group 1 (T1 Control, T2 Indoor nature)	21	85.7 (18)	19.5 (1.3)
Group 2 (T1 Indoor nature, T2 Control)	23	95.7 (22)	19.5 (1.3)
Group 3 (T1 Indoor nature, T2 indoor nature)	26	84.6 (22)	19.5 (1.6)
Study 2 – secondary school	**213**	**52.6 (112)**	**14.5 (1.4)**
Control	75	45.3 (34)	15.7 (1.0)
Potted Plants	73	52.7 (39)	13.5 (1.3)
Flowers	64	60.9 (39)	14.3 (0.8)
Study 3 – secondary vocational school	**161**	**62.7 (101)**	**17.8 (1.8)**
Control	43	58.1 (25)	17.7 (2.0)
Green walls	56	71.9 (41)	17.9 (1.8)
Potted plants	61	57.4 (35)	17.7 (1.8)

T1 measurement 1, T2 measurement 2.

The course leader of the course Research and Statistics randomly assigned students to one of the three included groups. The time, used materials, and content of the lectures were the same in all three classroom. Per group, the lectures at T1 and T2 were taught by the same teacher. The teachers were student-assistants who were equally experienced in teaching research and statistics subjects. Two researchers entered the classroom 45 min after the start of the lecture, and provided scripted instructions. Students were not informed about the exact objective of the study; they were told that the researchers were interested in students' well-being and the influence of the classroom. First, students completed a task to measure attention, and then students individually filled out a paper-based questionnaire. Completing the attention task and questionnaire took around 10 min.

2.1.2. Participants

Study 1 was conducted at the VU Amsterdam, the Netherlands. By means of convenience sampling, this study included first-year Bachelor Biomedical Science students enrolled in a course Research and Statistics. All 84 students that were present during T1 agreed to participate. At T2, 11 students were absent and one student declined to participate. At T2, two students violated protocol and followed the lecture in the wrong classroom, and thus with a different teacher. Teacher was a 'fixed variable' to prevent confounding, and therefore we excluded the two students who followed the lecture with a different teacher at T2. In total, 70 students were included in the statistical analyses. The study sample comprised 62 (88.6%) females, which reflects the typical gender distribution of Biomedical Science students in the Netherlands. Students' age ranged between 18 and 24 years with a mean age of 19.5 ($SD = 1.4$) (Table 1).

2.1.3. Stimuli

The three classrooms were located on the sixth floor on the north side of the same academic building ensuring comparable window views, light, and temperature in each classroom. Windows were closed and the blinds were open during the measurements. Each classroom had a maximum capacity of 30 students. Tables were set-up in long rows facing the blackboard. Nothing was changed in the control classroom. Experimental classrooms were decorated with indoor nature that consisted of various interior plants and green walls (125*200 cm) filled with plants of the species Philodendron Scandus (Fig. 1). We aimed to surround students with plants so that they would have a visual on nature from every angle. Plant species were selected in collaboration with suppliers and growers of interior plants based on plant robustness, the amount of natural light in the classroom, and aesthetics. Please see the supplementary materials for all the plants used and their details. The first classroom with indoor nature included two green walls in the front of the classroom next to the blackboard and 26 large and smaller plants, including Nepenthes species, Schefflera species, Spathiphyllum, Dracaena warneckeii, and Caryota mitis. Most of the plants were placed at the right side of the classroom in two niches in the front and rear corners of the classroom. Six small Nepenthes Aleta were placed in the windowsill. The second classroom included one green wall in the back of the classroom and 12 large and smaller plants, including Schefflera species, Dracaena surculosa, Kentia, Croton, Aglaonema, and Caryota mitis. Most plants were placed on the left side of the classroom in a small niche in the middle of the classroom, the front corner, and the rear corner of the classroom. Two plants were placed in the front right corner of the classroom.

2.1.4. Measurements

Attention was measured using the Digit Symbol Substitution Test (DSST) of the Wechsler Adult Intelligence Scale (WAIS) edition 3 [48,49]. At T1, a standardised version was used, and during the second measurement, a parallel-version was used [50]. The DSST was assessed in the classroom in 90 s using paper and pencil. The number of correct digits converted represents the magnitude of attention.

Well-being was measured using the sub-scales stress (i.e., tension), fatigue, and vigour of the Dutch Profile Mood States (POMS) [51]. Students were asked to indicate how they felt at the time of filling out the questionnaire. Students responded to the items on a five-point scale ranging from zero (not at all) to four (very much), and average scores were computed with higher scores reflecting more stress (α T1 and T2 = 0.8), more fatigue (α T1 and T2 = 0.8), and more vigour (α T1 = 0.8, α T2 = 0.9). Because the average scores of stress and fatigue showed a positive skew, towards very low scores, they were dichotomised around the overall sample median into no stress (scores 0–0.20) and somewhat to high stress (scores $\geq$0.21), and no fatigue (scores 0–0.60) and somewhat to high fatigue (scores $\geq$0.61).

Health complaints, during the lecture, were measured using nine items of the 12 item MM-questionnaire that were most likely to be affected by short exposure [52,53]: feeling heavy-headed, headache, nausea/drowsiness, concentration problems, itching/burning/irritation of the eyes, irritated/stuffy/running nose, hoarse/dry throat, a cough, and dry or flushed/red skin of the hand or face. Answer possibilities were on a four-point scale (not at all, sometimes, often, very often). An average score was created ranging from zero to three with higher scores reflecting more health complaints (α T1 = 0.7, α T2 = 0.6). Because this average score showed a positive skew, it was dichotomised around the overall sample median into no complaints (scores 0–0.44) and sometimes to often (scores $\geq$0.45).

Teacher evaluation was evaluated with four items adapted from a course evaluation questionnaire [54]. The items were: The teacher explained the materials clearly, encouraged students to think about the materials, made a clear distinction between main points and side issues, and adequately answered questions. Students responded to the four items on a five-point Likert scale ranging from strongly disagree to strongly agree. An average score ranging from zero to four was created with higher scores reflecting more positive teacher functioning (α T1 and T2 = 0.8).

Perceived environmental quality was evaluated with two measures. First, students were asked to give a numerical *grade* of the classroom on as scale from one (very poor) to ten (excellent). Second, *classroom attractiveness* was evaluated using an adaption of the environmental assessment scale [55]. The scale consisted of ten items representing adjectives: inspiring-drab or dull, attractive-ugly, peaceful-hectic, gloomy-cheerful, tacky-rich in nature. Students responded to the ten

Fig. 1. Classrooms included in Study 1: (A) control classroom (B) classroom with indoor nature 1 (C) classroom with indoor nature 2.

Table 2
Unadjusted descriptive statistics and results of longitudinal multi-level analyses (linear and binary logistic) adjusted for the three groups in Study 1 (N = 70).

	Control		Indoor nature		Intervention effect
	T1 *n* = *21*	T2 *n* = 23	T1 *n* = *49*	T2 *n* = *47*	
	Means (*SD*) or % (*n*)	Means (*SD*) or % (*n*)	Means (*SD*) or % (*n*)	Means (*SD*) or % (*n*)	B or OR (95% CI)
Attention	68.24 (10.29)	72.09 (9.89)	66.02 (9.52)	75.85 (9.19)	B = −0.84 (−3.53, 1.85)
Stress (somewhat to high)	42.9 (9)	39.1 (9)	44.9 (22)	44.7 (21)	OR = 1.00 (0.43, 2.43)
Fatigue (somewhat to high)	61.9 (13)	47.8 (11)	75.5 (37)	51.1 (24)	OR = 1.20 (0.52, 2.80)
Vigour	1.67 (0.80)	1.81 (0.77)	1.89 (0.73)	1.91 (0.85)	B = 0.13 (−0.11, 0.36)
Health complaints (sometimes to often)	33.3 (7)	34.8 (8)	51.0 (25)	21.3 (10)	OR = 0.90 (0.37, 2.21)
Teacher evaluation	2.58 (0.41)	3.09 (0.48)	3.06 (0.42)	2.76 (0.48)	B = 0.02 (−0.11, 0.15)
Classroom grade	6.62 (1.12)	6.56 (1.08)	7.41 (1.40)	7.55 (0.97)	B = 1.00 (0.64, 1.36)
Classroom attractiveness	1.53 (0.35)	1.46 (0.47)	2.11 (0.56)	2.09 (0.35)	B = 0.48 (0.38, 0.58)

B beta, OR odds ratio, T1 measurement 1, T2 measurement 2.

items on a four-point scale (not, a little, quite, very). After recoding the negative words, an average score was created ranging from zero to three with higher scores reflecting a higher attractiveness score (α T1 = 0.9, α T2 = 0.8).

2.1.5. Statistical analyses

Longitudinal multi-level analyses using MLwiN 2.36 software were performed to account for the clustering of measurements (level 1) within the individual students (level 2). The groups were added as a covariate to account for clustering within the three groups. The model only included a random intercept and no random slopes. The two classroom conditions were collapsed to estimate a general effect of indoor nature. Linear models were used for the dependent variables that were continues, and binary logistic models were used for dependent variables that were dichotomous.

2.2. Results

Table 2 shows the unadjusted means or proportions of the outcome variables and the results of the longitudinal multi-level analyses. There was a small difference in *attention* and *fatigue* favouring the control classroom, and a small difference in *vigour* and *health complaints* favouring the classroom with indoor nature. There were no noticeable differences in *stress* (OR = 1.00, 95% CI = 0.43, 2.43) and *teacher evaluation*. *Classroom grade* was 1 point (95% CI = 0.64, 1.36) higher in the classrooms with indoor nature, and *classroom attractiveness* was 0.48 points (95% CI = 0.38, 0.58) higher in the classrooms with indoor nature than the control classroom.

2.3. Conclusion

Supporting our hypotheses, students rated the perceived environmental quality higher in the classrooms with indoor nature. There were small differences in vigour and health complaints favouring the classroom with indoor nature, but their CIs were large. Mean scores on the attention task were higher at T2 than T1, which may reflect a maturation effect of the attention task. Students reported low levels of stress, fatigue, and health complaints; meaning that there is little room for interventions to improve these aspects.

3. Study 2

In Study 2, we examined a larger sample than in Study 1, and we carried it out in a different education setting. We included two follow-up measurements to explore short and longer-term effects, and included various types of indoor nature to assess the individual effects of flowers and potted plants versus the classroom standard design.

3.1. Method

3.1.1. Design and procedure

Initially, this study included four classroom conditions: a classroom with standard design (control), a classroom with a green wall, a classroom with potted plants, and a classroom with flowers. Due to loss to follow-up, there was no data available on the classroom with the green wall. This resulted in a three (classroom condition; between subjects) by three (measurements; within subjects) design. Measurements were administered between April and May 2017. Baseline measurements (T0) were conducted in April. Two week later indoor nature was placed, and three weeks later (the week directly after the introduction of indoor nature) the first follow-up (T1) was conducted. Four weeks after T1, a second follow-up (T2) was conducted. Measurements were on the same day of the week, on the same time, in the same classroom, and during the same lecture, being either Dutch, English, or French. Teachers were provided with instructions on how to explain the questionnaire to the students. The teacher provided the students with the paper (third and fourth-year students) or online (first and second-year students) questionnaire in the last 5 min of the 50-min lecture. Students individually filled out the questionnaires while still in the classroom. Students were told that the study was interested in students' well-being and the influence of the classroom.

3.1.2. Participants

The study was conducted at a secondary school that offers general secondary and pre-university education, in a small town (<5500 inhabitants) in the eastern part of the Netherlands. Dutch students are generally 12 years old when they start secondary education. By means of convenience sampling, 21 tutorial groups (*N* = 451) participated at baseline. Twenty-three students were excluded because their parents (*n* = 6) or they themselves (*n* = 17) did not provide informed consent. Three groups (*n* = 60) were excluded due to protocol violation. We excluded six groups because these groups did not fill-out the questionnaire at T1 or T2 (*n* = 152). These questionnaires were not filled-out because lectures were cancelled or because teachers did not offer the questionnaires to the students. We excluded these groups because our statistical models were adjusted for baseline values. This means that, if we included these six groups, the statistical models had to predict estimates based on only one measurement (i.e., T1 of T2). Given that the data within these groups were correlated, this would have led to over- or underestimations. Of the remaining students (*n* = 217), four were absent during both T1 and T2 and, therefore, excluded. The statistical analyses included 213 students. The study sample comprised 112 females (52.6%), and age ranged between 12 and 18 years with a mean of 14.53 years (*SD* = 1.42). Because of exclusion of complete groups, there were no third-year students in the control classroom, no first and fourth-year students in the classrooms with plants, and no first and second-year students in the classrooms with flowers.

3.1.3. Stimuli

The classrooms were located on the ground floor on the south side of the school building ensuring comparable window views, light, and temperature in each classroom. Windows could not be opened, and there was no record if the blinds were open or closed during the measurements. Each classroom had a maximum capacity of 30 students. Tables were set-up in pairs facing the blackboard and the teachers' desk. Nothing was changed in the control classroom. There were two experimental conditions: a classroom with potted plants and a classroom with flowers (Fig. 2). The classroom with plants included five large pots with various plant species, which were selected in collaboration with suppliers and growers of interior plants based on plant robustness, the amount of natural light in the classroom, and aesthetics. Plant species included Philodendron Selloum, Schefflera Arbocicola, Calathea Sanderiana, and Zamioculcas Zamiifolia (more details in the supplementary materials). Because the density of plants was smaller than in Study 1, we tried to make sure that the plants were visual when students entered the room and when they were facing the blackboard. Two plants were placed in the front of the classroom next to the teachers desk in the windowsill, one plant was placed in the front of the classroom on a cabinet, and two plants placed in the back of the classroom on another cabinet. In the classrooms with flowers, a large field bouquet in a vase was placed in the front of the classroom behind the desk of the teacher. Nine small vases each holding one rose were placed in the front corner of the classroom on a small cabinet. Seven small vases each holding one rose were placed at the back corner of the classroom. The roses had been especially cultivated to have a stronger scent than regular roses. Fresh flowers were placed every Monday morning.

3.1.4. Measurements

Well-being was again measured with the sub-scales stress (T0 $\alpha = 0.8$, T1 and T2 $\alpha = 0.9$) and fatigue (T0, T1 and T2 $\alpha = 0.9$) of the Dutch POMS [51], and thereafter dichotomised using the same thresholds as in Study 1. *Health complaints* (T0, T1, and T2 $\alpha = 0.8$) were again measured with the MM-questionnaire, yet fatigue was added as an extra symptom [52,53]. The scale was dichotomised using the same thresholds as in Study 1. *Teacher evaluation* (T0 $\alpha = 0.8$, T1 and T2 $\alpha = 0.9$) and *classroom grade* were evaluated with the same scales as in Study 1. *Classroom attractiveness* (T0 $\alpha = 0.8$, T1 and T2 $\alpha = 0.9$) was also measured with the same scale as in Study 1. Answer options were changed to a five-point Likert scale, ranging from strongly disagree to strongly agree.

In addition to these scales, self-reported *attention* was measured using three items that were based on items of sub-scales of the Checklist Individual Strength [56] and the Amsterdam Executive Function Inventory [57]. The items were: I could not focus on the same topic for a long period, I was easily distracted, and my thoughts easily wandered. Students responded to the items on a three-point scale (not true, partly true, true). After recoding, an average score was created ranging from zero to two with higher scores reflecting better attention (T0, T1 and T2 $\alpha = 0.9$). *Lecture evaluation* was measured with five items from a course evaluation questionnaire [54]. An example item: It was an interesting lecture. Students responded to the five items on a five-point Likert scale ranging from strongly disagree to strongly agree. Average scores (T0, T2 $\alpha = 0.8$ and T1 $\alpha = 0.9$) were created ranging from zero to four with higher scores reflecting more positive lecture evaluation.

3.1.5. Statistical analyses

Similar to Study 1, longitudinal multi-level analyses were used to account for the clustering of measurements within students. The inclusion of a third level, the tutorial group, was considered and discarded based on likelihood ratio tests in the linear models [58]. The statistical models included the classroom condition, measurements (T1 and T2), and interactions between measurement and classroom condition to estimate the intervention effects at T1 and T2. Baseline scores of the dependent variables were added to the model to correct for existing differences between the classrooms at baseline [59].

3.2. Results

Table 3 shows the unadjusted means or proportions of all outcome variables and the results of the longitudinal multi-level analyses. *Attention* at T1 was higher in the classroom with plants than in the control classroom (B = 0.23, 95% CI = 0.03, 0.42). There was no difference between the classroom with flowers and the control classroom. At T2, attention was higher in both classrooms with indoor nature than the control classroom. Students had lower odds of reporting somewhat to high *stress*, somewhat to high *fatigue*, and sometimes to often *health complaints* in both classrooms with indoor nature than those in the control classroom at T1 and T2. *Lecture* and *teacher evaluations, classroom grade*, and *classroom attractiveness* were higher in both classrooms with indoor nature than the control classroom at T1 and T2.

3.3. Conclusion

Students who followed a lecture in the classrooms potted plants or the classroom with flowers reported greater attention, more positive lecture evaluation, and more positive environmental quality than those in the control classroom. The results suggest that these effects were not based on novelty because they favoured the classrooms with indoor nature at T1 and T2. Teacher evaluations favoured the classrooms with indoor nature at T1 and T2; however, at T2 these effects were smaller. Measures of stress, fatigue, and health complaints also favoured the classrooms with indoor nature over the control classroom, but large CIs were found. This might, similar to Study 1, again be the result of floor effects. At T2, a larger and more precise effect on fatigue in the classroom with plants was found. Nevertheless, this might be attributed to the increase in fatigue in the control group from baseline to follow-up, because the proportions of fatigue in the classroom with plants remained same over time. Despite that flowers showed potential benefits, they also turned out to be less practical for implementation. Flowers need to be replaced weekly and their freshness deteriorates over time, whereas plants can keep their freshness for a longer period.

4. Study 3

In Study 3, we included green walls because we were unable to

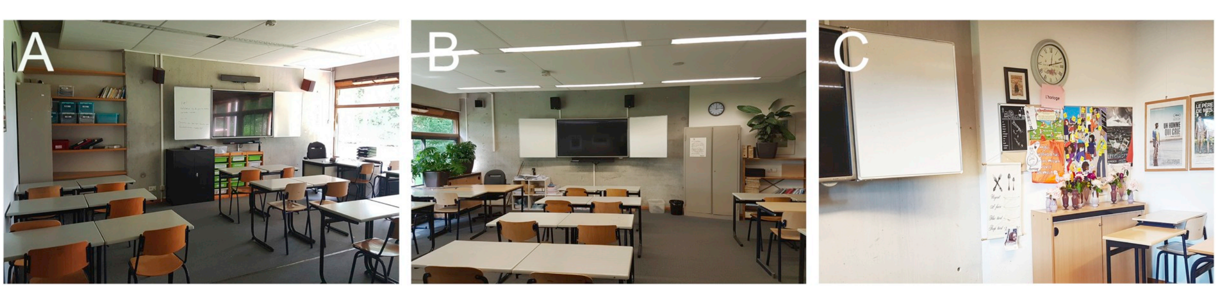

Fig. 2. Classrooms included in Study 2: (A) control classroom (B) classroom with potted plants (C) classroom with flowers.

Table 3
Unadjusted descriptive statistics and results of longitudinal multi-level analyses (linear and binary logistic) adjusted for baseline values in Study 2 (N = 213).

		T0	T1	T2	Intervention effects T1	Intervention effects T2
		Means (*SD*) or % (*n*)	Means (*SD*) or % (*n*)	Means (*SD*) or % (*n*)	B or OR (95% CI)	B or OR (95% CI)
Attention	Control (ref)	1.02 (0.71)	1.14 (0.63)	1.04 (0.64)		
	Potted plants	1.60 (0.45)	1.55 (0.57)	1.49 (0.60)	B = 0.23 (0.03, 0.42)	B = 0.23 (0.05, 0.41)
	Flowers	1.35 (0.61)	1.28 (0.65)	1.53 (0.62)	B = 0.01 (−0.19, 0.20)	B = 0.35 (0.17, 0.53)
Stress (somewhat to high)	Control (ref)	32.0 (24)	32.0 (24)	40.0 (30)		
	Potted plants	33.8 (25)	41.9 (31)	39.1 (25)	OR = 0.91 (0.40, 2.07)	OR = 0.70 (0.32, 1.51)
	Flowers	37.5 (24)	40.6 (26)	33.8 (25)	OR = 0.80 (0.34, 1.88)	OR = 0.89 (0.40, 1.99)
Fatigue (somewhat to high)	Control (ref)	66.7 (50)	49.3 (37)	69.3 (52)		
	Potted plants	43.2 (32)	41.9 (31)	40.5 (30)	OR = 0.47 (0.19, 1.14)	OR = 0.34 (0.15, 0.77)
	Flowers	46.9 (30)	46.9 (30)	53.1 (34)	OR = 0.61 (0.24, 1.51)	OR = 0.68 (0.29, 1.61)
Health complaints (sometimes to often)	Control (ref)	54.7 (41)	37.3(28)	50.7 (38)		
	Potted plants	39.2 (29)	33.8 (25)	32.4 (24)	OR = 0.52 (0.22, 1.25)	OR = 0.52 (0.23, 1.15)
	Flowers	50.0 (32)	32.8 (21)	29.7 (19)	OR = 0.42 (0.17, 1.05)	OR = 0.37 (0.16, 0.88)
Lecture evaluation	Control (ref)	1.97 (0.64)	1.66 (0.86)	2.03 (0.86)		
	Potted plants	2.54 (0.59)	2.49 (0.63)	2.53 (0.53)	B = 0.58 (0.33, 0.82)	B = 0.23 (0.00, 0.45)
	Flowers	2.20 (0.77)	2.52 (0.65)	2.49 (0.86)	B = 0.78 (0.53, 1.03)	B = 0.36 (0.13, 0.59)
Teacher evaluation	Control (ref)	2.04 (0.86)	1.81 (0.94)	1.90 (0.91)		
	Potted plants	2.83 (0.55)	2.77 (0.70)	2.56 (0.66)	B = 0.48 (0.21, 0.75)	B = 0.23 (−0.02, 0.47)
	Flowers	2.50 (0.93)	2.71 (0.67)	2.24 (1.07)	B = 0.60 (0.33, 0.87)	B = 0.10 (−0.15, 0.34)
Classroom grade	Control (ref)	5.18 (1.67)	5.25 (1.59)	4.85 (1.78)		
	Potted plants	6.23 (1.31)	7.02 (1.10)	6.82 (1.14)	B = 1.21 (0.80, 1.63)	B = 1.36 (0.97, 1.75)
	Flowers	6.68 (1.20)	6.54 (1.32)	6.60 (1.40)	B = 0.47 (0.02, 0.93)	B = 0.92 (0.49, 1.34)
Classroom attractiveness	Control (ref)	1.18 (0.56)	1.34 (0.55)	1.31 (0.67)		
	Potted plants	1.59 (0.67)	2.22 (0.58)	2.00 (0.66)	B = 0.66 (0.47, 0.84)	B = 0.42 (0.25, 0.59)
	Flowers	1.85 (0.63)	2.07 (0.71)	2.04 (0.62)	B = 0.32 (0.12, 0.52)	B = 0.33 (0.14, 0.52)

B beta, OR odds ratio, T0 baseline, T1 follow-up 1, T2 follow-up 2.

measure their effects in Study 2. We did not include flowers due to their impracticality.

4.1. Method

4.1.1. Design and procedure

Similar to Study 2, Study 3 included a three (classroom condition; between subjects) by three (measurements; within subjects) design. Classrooms conditions included a classroom with standard design, a classroom with potted plants, and a classroom with green walls. Measurements were administered at baseline (T0), two weeks after introduction of indoor nature in the classrooms (three weeks after baseline; T1), and five weeks after the introduction of indoor nature in the classrooms (T2). Data collection took place in November and December 2017 on Mondays between 08:30 and 14:15. Measurements were in the same classroom and during the same courses, being either Dutch, English, or Biology. Guided by scripted instructions, one researcher and an assistant administered a task to measure attentional fatigue at the beginning and the end of the 50-min lecture. At the end of the lecture, questionnaires were also completed. Students were told that the researchers were interested in students' well-being and the influence of the classroom.

4.1.2. Participants

Study 3 was conducted at a secondary vocational school located in a midsized town (+- 31,430 inhabitants) in the northern part of the Netherlands. Secondary vocational education generally follows after students finish their pre-vocational secondary education at the age of 16 years, and it prepares students for various occupations depending on the level of training [60]. This school offers vocational training on training levels one to four with courses on animal welfare and horticulture related courses. Twelve groups were included at baseline ($N = 189$). Seventeen students were excluded because they did not give informed consent at one of the three measurements. Students that were absent during both T1 and T2 were also excluded ($n = 11$). As a result, 161 students were included in the statistical analyses. The study sample comprised 101 (63%) females, and their age ranged between 15 and 27 years with a mean of 17.8 ($SD = 1.8$) years. At this school, some courses have a small number of students. Therefore, transcending lectures, such as Dutch, English, and Biology were taught in groups that consisted of students from various courses or levels. Consequently, the classroom with plants did not include students from the third and fourth-year, and the classroom with green walls included only students from training level four (the highest level) whilst the classroom with plants did not include students from level four.

4.1.3. Stimuli

The three classrooms were located on the first floor of the same education building, oriented at the same side of the building ensuring comparable window views, light, and temperature in each classroom. Windows were closed during T0 and T2, and open in each classroom during T1. The blinds were open during each measurement. Tables were set-up in pairs facing the blackboard and the teachers' desk. Two months before the baseline measurement, the walls of all three classrooms were painted white. After that, nothing was changed in the control classroom. There were two experimental conditions: a classroom with potted plants and a classroom with green walls (Fig. 3). Plant species were selected in collaboration with suppliers and growers of interior plants based on plant robustness, the amount of natural light in the classroom, and aesthetics. Please see the supplementary materials for the plant species used and their details. The classroom with the potted plants included 17 large and smaller plants, such as Codiaeum species, Dracaena species,

Fig. 3. Classrooms included in Study 3: (A) front-side classroom with potted plants (B) back-side classroom with potted plants (C) classroom with green walls.

Sansevieria, and Dieffenbachia. The plants were placed in four different places in the classroom in order to create different plant islands based on colour. One red coloured wall plate and one green coloured wall plate was placed behind two of the plant islands located in the back of the classroom. In the front of the classroom, the other two plant islands were placed. Platforms were used to create a variation of heights making sure that the plants were on eye level. A total surface area of approximately 10 m^2 green was used. The classroom with the green walls (125*200 cm) included four planters filled with plants of the species Philodendron Scandus. A total surface area of approximately six m^2 indoor green was used.

4.1.4. Attention test and questionnaire measurements

Based on a pilot test among 30 vocational school students, the measures used in the first two studies were simplified and supported by emoticons of thumbs or smiley faces. Similar to Study 2, self-reported *attention* was measured with three items. However, "I could not focus on the same topic for a long period" was simplified to "I could concentrate well", based on items of the concentration sub-scale of the Checklist Individual Strength [56]. Answer possibilities were on a three-point scale (no, a little, yes). Because the average score (T0, T1, and T2 $\alpha = 0.8$) was negatively skewed, attention was dichotomised around the median into poor attention (scores <1) and good attention (scores of $\geq$1). The *classroom grade* and *attractiveness* (T0 $\alpha = 0.7$, T1 and T2 $\alpha = 0.9$) were similar to Study 2 and supported by illustrations.

The DSST was used to measure *attentional fatigue* [48,49]. At the beginning of class, the WAIS edition 3 was used and at the end of class, the WAIS edition 4 was used. The DSST was administered in the classroom by two researchers. Duration of the test was 90 s, and students filled-out the test using paper and pencil. Attentional fatigue was calculated as the difference between the number of correct digits converted before and after class. A negative score represents less correct digits converted after the lecture, an indicator of attentional fatigue. The results of the DSST at T2 of two groups were eliminated due to errors in the time tracking.

Items previously used by a study into restorative effects of green walls in classrooms were used to assess *mood* [41]. Mood after the lecture was assessed on four emotions (joy, happiness, fatigue, relaxed). These four emotions were assessed on a five-point Likert scale ranging from not at all applicable to very applicable. Average scores were created with higher scores reflecting more positive mood (T0 $\alpha = 0.6$, T1 and T2 $\alpha = 0.7$).

Health complaints were measured using four items adapted from the health complaints MM-questionnaire (headache, nausea, a cough, dry or sore throat) [52,53]. The frequency of complaints during the lecture was rated on a three-point scale (no, little, yes). An average sum score was created ranging from zero to two with higher scores reflecting more complaints (T0 and T2 $\alpha = 0.7$, T1 $\alpha = 0.6$). As this scale was positively skewed, it was dichotomised around the median into no complaints (scores 0) and any health complaints (scores >0).

4.1.5. Statistical analyses

The questionnaire and attentional fatigue data were analysed in the same way as in Study 2: longitudinal multi-level models with two levels (time within participant) with a correction for baseline values.

4.2. Results

Table 4 shows the unadjusted means or proportions for the outcome variables and the results of the longitudinal multi-level analyses. *Attentional fatigue* ranged from −31 to 38. At T0, average scores were negative indicating lower scores on the attention test after the lecture than before the lecture. At T1 and T2, average scores were positive. At T1, students in the classrooms with indoor nature showed more attentional fatigue than those in the control classroom. At T2, these effects were reversed. *Attention* was lower in both classrooms with indoor nature than the control classroom at T1 and T2, except for the classroom with plants at T2. At both T1 and T2, *mood* was higher in both classrooms with indoor nature than in the control classroom, except at T2 mood was 0.07 points lower (95% CI = −0.29, 0.15) in the classroom with plants than the control classroom. Students had higher odds on reporting any *health complaints* at T1 and T2 in the classrooms with indoor nature than the control classroom, but there was no difference between the classroom with plants and the control classroom at T2. *Classroom grade* and *attractiveness* were rated higher in both classrooms with indoor nature than the control classroom at T1 and T2.

4.3. Conclusion

Similar to the first two studies, Study 3 showed that indoor nature in classrooms contributes to more positive classroom attractiveness scores. Classroom grades were slightly higher in the classrooms with indoor nature, but their CIs were large. Only in the classroom with the green walls at T1, a more precise and larger effect on classroom grades was found. No meaningful intervention effects were found for attentional fatigue and the other self-reported measures: the differences were small and CIs large.

5. General discussion

In three field experiments conducted at a university, a secondary school, and a secondary vocational school, we explored the effects of various types of indoor nature in classrooms after one lecture. Consistent with our hypotheses, results showed that students preferred classrooms with indoor nature to the control classrooms. Secondary school students also reported greater attention, lecture evaluations, and teacher evaluations in the classrooms with indoor nature compared to those without. Contrary to our hypotheses, no meaningful effects on well-being and health complaints were found. In Study 2 and 3, we explored various types of indoor nature, which seemed to have comparable effects.

As noted in the introduction, the beneficial effects of nature might be explained by the biophilia hypothesis and theories focusing on nature's contribution to restoration. Consistent with the biophilia hypothesis, which holds that all humans innately respond positively to natural elements [14], students seem to prefer classrooms with indoor nature over those without because they were rated higher on environmental quality.

Table 4

Unadjusted descriptive statistics and results of longitudinal multi-level analyses (linear and binary logistic) adjusted for baseline values in Study 3 (N = 161).

		T0	T1	T2	Intervention effects T1	Intervention effects T2
		Means (SD) or % (n)	Means (SD) or % (n)	Means (SD) or % (n)	B or OR (95% CI)	B or OR (95% CI)
Attentional fatigue	Control (ref)	−1.53 (7.01)	5.38 (10.11)	−0.04 (10.56)		
	Green walls	−2.35 (6.33)	3.52 (8.29)	1.19(8.03)	B = −1.10 (−5.09, 2.89)	B = 2.35 (−2.73, 7.44)
	Potted plants	−1.15(7.24)	2.13 (4.68)	4.02 (11.84)	B = −3.04 (−6.85, 0.78)	B = 3.40 (−0.70, 7.50)
Attention (good)	Control (ref)	53.5 (23)	60.5 (26)	48.8 (21)		
	Green walls	57.9 (33)	61.4 (35)	42.1 (24)	OR = 0.99 (0.36, 2.70)	OR = 0.76 (0.28, 2.09)
	Potted plants	67.2 (41)	49.2 (24)	59 (36)	OR = 0.48 (0.18, 1.27)	OR = 1.13 (0.42, 3.03)
Mood	Control (ref)	2.33 (0.68)	2.23 (0.57)	2.19 (0.75)		
	Green walls	2.23 (0.64)	2.34 (0.62)	2.21 (0.51)	B = 0.12 (−0.10, 0.34)	B = 0.10 (−0.12, 0.33)
	Potted plants	2.24 (0.54)	2.25 (0.57)	2.07 (0.69)	B = 0.06 (−0.16, 0.28)	B = −0.07 (−0.29, 0.15)
Health complaints (any)	Control (ref)	41.9 (18)	34.9 (15)	48.8 (21)		
	Green walls	38.6 (22)	36.8 (21)	52.6 (30)	OR = 1.15 (0.48, 2.77)	OR = 1.23 (0.53, 2.87)
	Potted plants	57.4 (35)	57.4 (35)	52.5 (32)	OR = 2.24 (0.95, 5.29)	OR = 0.96 (0.41, 2.21)
Classroom grade	Control (ref)	6.49 (1.14)	6.33 (0.98)	6.35 (1.12)		
	Green walls	6.24 (1.36)	6.96 (1.20)	6.66 (1.12)	B = 0.71 (0.25, 1.17)	B = 0.46 (−0.01, 0.94)
	Potted plants	6.53 (1.37)	6.71 (1.41)	6.51 (1.37)	B = 0.35 (−0.10, 0.80)	B = 0.19 (−0.27, 0.65)
Classroom attractiveness	Control (ref)	1.94 (0.49)	1.69 (0.63)	1.85 (0.66)		
	Green walls	1.51 (0.48)	2.16 (0.56)	2.09 (0.51)	B = 0.65 (0.42, 0.87)	B = 0.44 (0.21, 0.67)
	Potted plants	1.69 (0.53)	2.29 (0.55)	2.25 (0.58)	B = 0.67 (0.45, 0.89)	B = 0.51 (0.29, 0.73)

B beta, OR odds ratio, T0 baseline, T1 follow-up 1, T2 follow-up 2.

These results were robust even though various types of indoor nature across the three experiments were used. It seems that these results were also not based on novelty because Study 2 and 3 showed that students still preferred the classrooms with indoor nature to the control classroom a few weeks after indoor nature was placed. These findings are compatible with a study among Dutch university students, who preferred simulated study environments with some sort of nature over those without or with colourful murals [61]. Preference for classrooms with indoor nature has also been found in pupils [41,44]. Incorporating indoor nature can thus improve students satisfaction with their study environment, which in the long term may positively influence retention and students' beliefs about their academic performance [62,63].

In the perspective of nature's contribution to restoration [6,20–22], we expected that indoor nature would contribute to better performance on attention tasks and improve self-reported attention, well-being, and teacher and lecture evaluations. With respect to attention, only secondary education students reported greater attention in the classrooms with indoor nature compared to the control classroom. In the other two studies, no meaningful differences on the attention tasks or self-reported attention was found. Attention measures differed between the three studies, which might resulted in dissimilarities between the three studies. Nevertheless, field experiments in primary school classrooms and offices have shown better scores on attention tasks, self-reported attention, and productivity [12,17,41]. Experimental research showed that micro-breaks of 40 s looking at green roofs can already boost attention in university students [64]. In our work, we did not find comparable effects. This may be the result of several complementary factors. First, Ohly et al. [65] proposed that the Digit Symbol Substitution test may not be appropriate to test attention restoration. They also suggest that a better understanding of attention restoration, and the tools to measure this, is needed [65]. Second, exposure to indoor nature does not mean that students engaged with nature in our experiments. It might be that he classroom setting does not allow for restorative experiences. Third, it may be that one lecture alone does not call upon the direct attention resources as much that attention restoration is needed.

With respect to well-being, effects generally favoured indoor nature to the control classroom. However, effects were small and CIs large. Stress and fatigue were included at the secondary school and university as measures of well-being. In both studies, students reported very low levels of stress and fatigue from the onset. With such floor effects, there is little room for interventions to improve these aspects. Restorative experiences are most likely and effective if there is a need for recovery [21]. Mood was included as a measure of well-being at the secondary vocational school. Similar to our work, previous studies have found no effects of indoor nature in university and primary school classrooms on mood [41,47]. However, laboratory studies on the effects of indoor nature have shown improved mood [23,66] and reduced stress [24]. Although laboratory studies have many advantages, their results may not relate to real settings. Most laboratory studies examine the effects of nature on restorative effects, logically, first induced stress [22,67], or those focusing on the cognitive benefits often use various tasks to simulate a cognitive load [24,64,66,68]. It might be that such simulations do not reach the levels of stress and cognitive load to the same degree that students experience while present in the classroom. It would be interesting to consider the ecological validity of future laboratory studies by, for example, manipulating induced stress. Future field experiments should explore the effects of indoor nature in different ways to further the understanding of its contribution to well-being. For example, with behavioural observations or objective measures (e.g. grades, sick leave).

With respect to lecture and teacher evaluations, secondary education students rated their lecture and teacher more positively in the indoor nature classrooms. University students reported no difference in teacher evaluation between the classrooms. Field experiments that conducted their follow-up at the end of a several weeks course found more positive evaluations of the course and professor in classrooms with plants compared to no plants [46], and classrooms with window nature views compared to no window views [69]. Improved lecture evaluations may reflect of enhanced environmental quality, feelings of comfort, or emotional states in both the teacher as the students. However, the latter is not supported by our work.

In our work, no differences in health complaints were found. Other studies have shown that indoor nature is associated with reduced health complaints in pupils and office workers [44,53] and sick leave in (junior) high school students [43]. Not finding effects on health complaints, however, does not mean that indoor nature cannot contribute to students' physical health. Plants have been shown to have a positive effect on the indoor climate, including increasing the relative humidity and reducing temperature, levels of carbon dioxide, volatile organic compounds, and particular matter [38,39,70]. By improving the indoor climate, plants may improve human comfort and health. Previous studies [36,43,44,53] measured physical health after a daily exposure over several weeks. In our experiments, we tried to capture similar effects after one lecture. Students reported very few health complaints after one lecture. It is thus possible that our exposure (one lecture) was too short to improve physical health and that our health complaints scale was unsuitable for our design.

5.1. Strengths and limitations

Our work has several strengths. The three studies included within-subject measures, and the results of Study 2 were based on a large sample. Results of Study 2 and 3 were adjusted for baseline differences. The classrooms included in this work were selected based on their comparability on aspects such as orientation, window views, and capacity. This ensured that the findings were not biased by classroom characteristics other than the indoor nature. Additionally, the studies were conducted in real-life classroom settings, which ensure high ecological validity.

The high ecological validity did come at the expense of many uncontrollable confounding factors. The classroom social climate or incidents during the lectures are examples of aspects we could not account for. We further had real-life problems such as students or teachers that were absent. This lead to unequal distributions of subgroups based on education year and training level in Study 2 and 3. Furthermore, in these two studies, we included lectures that varied within and between classroom conditions of which we assumed had similar impacts on the students. To overcome confounding issues in field experiments together with the small effect sizes that can be expected from the impact of indoor nature, a large sample size is needed to obtain sufficient statistical power. It is also important to bear in mind that the use of questionnaires is always subject to response bias, and that participating in experiments may cause demand characteristic bias or Hawthorne effects [71]. However, the latter was controlled for by adding a control group because the magnitude of such confounding effects was expected to be equal in the control group and experimental groups. The generalisability of the findings of Study 3 may also be subject to limitations because we also included students who followed horticulture related courses. These students may be exposed to nature more often due to their profession and interests. As a result, these students may experience less awe by indoor nature. This may be reflected in the classroom grades because the students in Study 3, following vocational animal welfare and horticulture related education, reported lower classroom grades for the classrooms with indoor nature than the university students and secondary education students in Study 1 and 2.

5.2. Conclusion

To conclude, the findings of the three field experiments only partly support our hypotheses. Findings consistently showed that students prefer classrooms with indoor nature to classrooms with the standard design. In Study 2, there were indications that attending a single lecture in a classroom with indoor nature contributes to students' self-reported attention, lecture evaluation, and teacher evaluation. However, attending a short period in a classroom with indoor nature does not seem to have immediate positive effects on students' well-being and health complaints. It is also noteworthy that all students reported low levels stress, fatigue, and health complaints. This may imply that after a single lecture, students do not have a high need to unwind, relax, or restore. As such, students may benefit more from nature in the education setting at times and places when there is a higher need to unwind, relax, or restore, or from indoor nature at places that are used for recovery activities such as study break spaces. Moreover, the effects of indoor nature on students' attention and well-being are likely to be influenced by many contextual factors. Replication under other circumstances is therefore needed.

Role of funding source

This work was supported by the Topsector Horticulture and Starting materials (kv-1604-050), with co-funding of the Triodos Foundation and Royal Floraholland. The funding sources were not involved in the development, execution, or interpretation of the results of the three experiments.

Ethical standards disclosure

The design of experiments two and three were approved by the Scientific and Ethical Review Board of the Faculty of Behavioral and Movement Sciences of the Vrije Universiteit Amsterdam, the Netherlands (vcwe-2017-032r1, vcwe-2017-128).

Authorship

NVDB, SCD, KTD, JCS, SLK, and JM were involved in the set-up of the study designs and materials. NVDB was responsible for data collection, statistical analyses, and the first draft of the manuscript. All others authors made substantial contributions with regard to interpretation of the results and to the draft manuscript. NVDB subsequently prepared a final version of the manuscript, which was approved by all authors for submission.

Declaration of competing interest

The indoor nature used in the three experiments was provided by various suppliers on a loan-base. These suppliers were not involved in the development, execution, or interpretation of the results of the three experiments. The authors have no other conflicts of interest to declare.

Acknowledgements

We would like to thank Kees Janssen, Martin Hogenboom, Nico Wissing, Gerrit Roessink, and Bart Bresser for their practical input with choosing and advising on plant species and designing the indoor nature classrooms. We also like to thank Tom Beugels from Cloud Garden for his assistance with the indoor climate sensors. We also express our gratitude to all students and colleagues who helped us with data collection or data entering: Eveline Draaisma, Babette Kales, Fabienne Mokken, Laura van de Harst, Philinne Mol, Liset Elsgeest, Janke van Dijk-Wesselius, Simon Duindam, Koen Aartsen, Corine van Hooven, Ibtissama Moussa, Yasemin Tunbul, Frédérique Rongen, and Daphne Meuwese.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.buildenv.2020.106675.

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